



Constellation

Constellation Energy

Recommendation: BUY

Lukas Wehlitz, Anton Motiashov, Nicholas Hortatsos, Evan Ross, Brady Krmpotich, Magnus Schoville, and Sam Pliner

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Recommendation: **BUY**

Timeline: 24+ Months

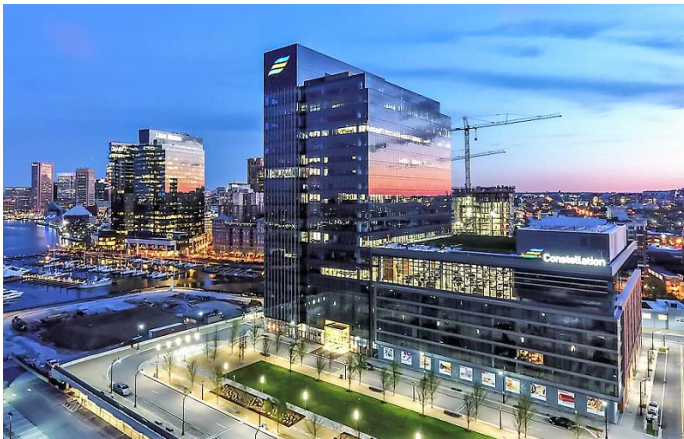


Company Information

- **Company Name:** Constellation Energy Corp.
- **Current Price:** \$232.28 (11/4 close)
- **Target Price:** \$365.09
- **Implied Return:** 53%
- **Thesis Points:**
 - 1) Investment in new plants will result in higher future cash flows
 - 2) Increasing demand for energy will increase demand for nuclear energy
 - 3) CEG is ideally positioned to capture growing demand in clean energy market

Key Information

- Spun off from Exelon in February 2022, when it opened at \$48.73
- Headquartered in Baltimore, Maryland
- Their goal is to contribute to the decarbonization of the energy sector while maintaining reliable affordable energy for their consumers
- 23 nuclear power plant units across multiple sites
- Powers more than 16 million homes and businesses
- Providing 10 percent of all clean power on the grid in the U.S.



Annotated Stock Chart

Last 24 Months

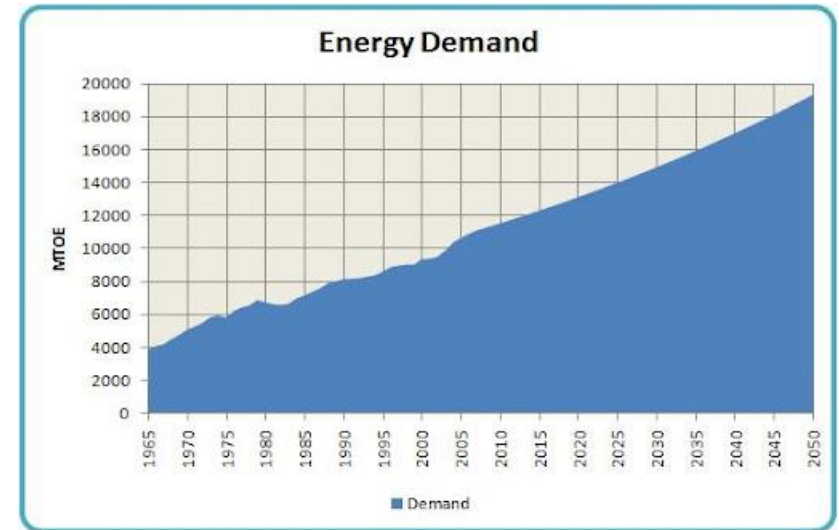
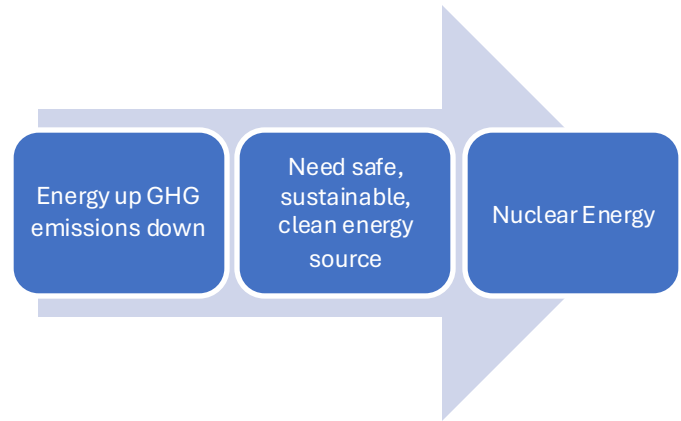
Key Changes in Share Price



- 1) **December 2023:** CVS announces deal with CEG that totals 264,000 megawatt hours of energy
- 2) **April 2024:** AAI gives CEG a value grade of D
- 3) **July 2024:** Signs a 20-year power purchase agreement with Microsoft
- 4) **November 2024:** Stock falls after US rejects Amazon power deal

Energy and Utilities Sector

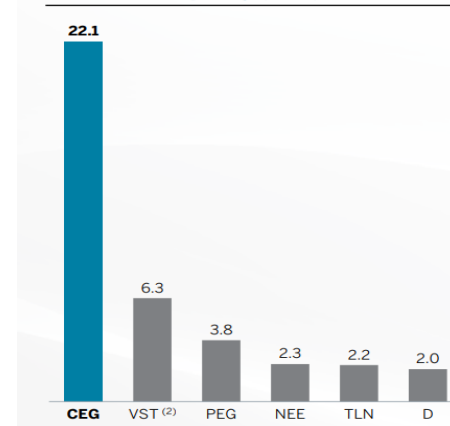
- Both make and distribute energy
- Non-renewable vs Renewable energy
- Highly regulated
- Energy demands increasing every year, but need to meet demands while being environmentally friendly
- Electrification of the U.S.
 - "Nuclear energy is currently largest source of zero emissions electricity in the U.S., accounting for over 50% of the nation's carbon-free power"



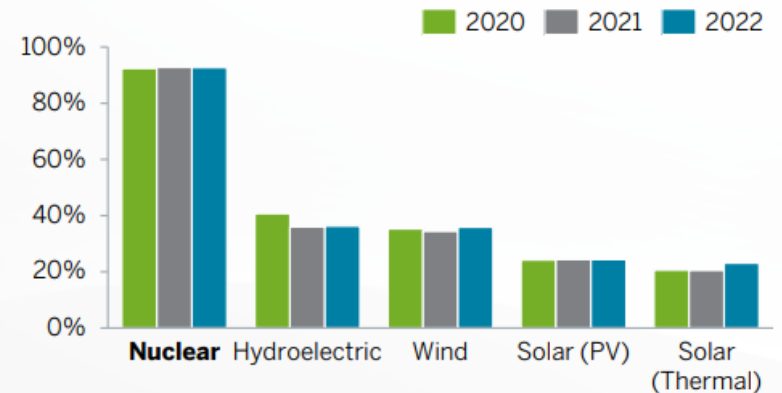
Competitive Landscape

- High initial investment costs cause nuclear energy market to not be very competitive or saturated
- Main competitors: Duke Energy and NextEra Energy
- Competition against alternative energy sources
 - "Advancement in new power generation technologies like residential solar generation and micro-turbine installations"

Competitive Nuclear Generation Capacity (GWs) ⁽¹⁾



Highest Capacity Factors ⁽²⁾





Thesis Point 1 – Investment in new plants will result in higher future cash flows

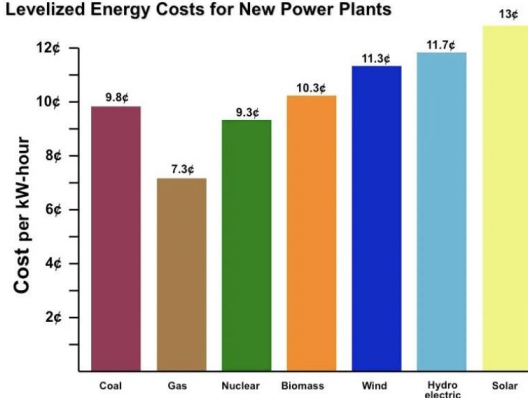
First Thesis Point

Investment in new plants will result in higher future cash flows

Nuclear Energy: Low Operational Costs

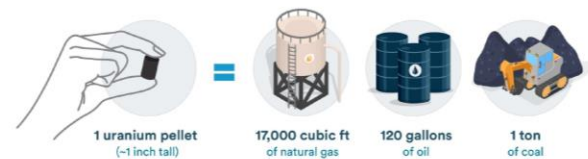
- Initial investment in nuclear energy is expensive but operational costs are lower compared to other forms of clean energy
- Low operational costs:
 - Uranium is energy dense which means low fuel costs
 - Nuclear energy is reliable (high capacity factor)
 - Stable operational costs enable the initial investment to be amortized over a long period
 - Production Tax Credit (PTC) and Zero-Emission Credit (ZEC)

Levelized Energy Costs for New Power Plants



The Levelized Cost of Energy for new power plants from various energy sources showing that for new construction, natural gas and nuclear are the two cheapest sources of electricity generation in the near-future. Data from IER 2015 Report [-]

Nuclear Fuel is Dense ⁽¹⁾



A Nuclear Plant's Footprint is Small ⁽²⁾



A 1,000 MW nuclear facility needs just over one square mile



A wind farm of similar generation output requires up to 360 times the land area



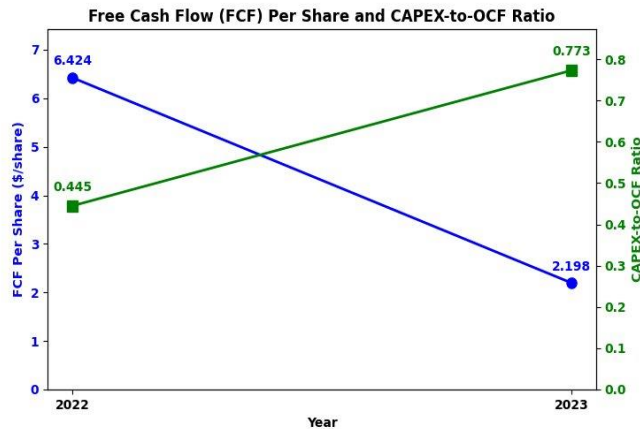
A solar (PV) facility of similar generation output requires up to 75 times the land area

First Thesis Point Continued

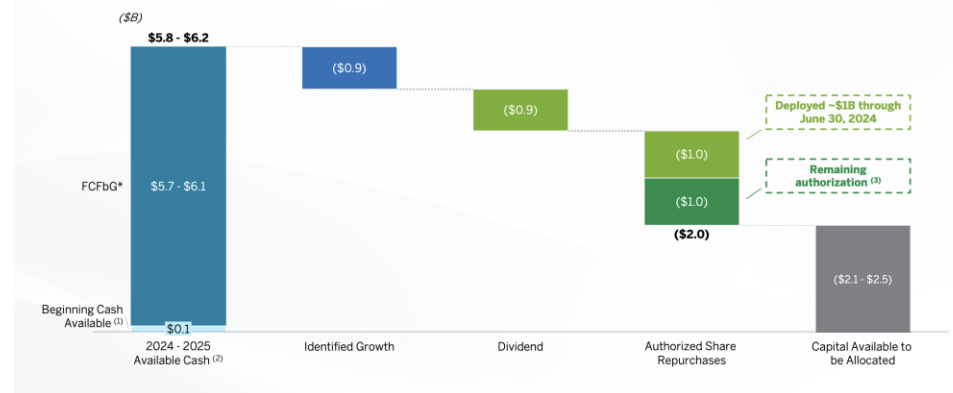
Investment in new plants will result in higher future cash flows

Low Operational Costs Lead to Higher Cash Flows

- Increase in CAPEX-to-OCF ratio shows a shift to a strong investment in CEG's future
 - E.g. Three Mile Island Nuclear Plant
- Still \$2.3 billion of capital to invest in new plants
- Low operational costs after initial high CAPEX will lead to higher future FCF
- FCF per share growth is strongly correlated with share price growth



Approximately \$2.3 Billion of Capital Still to Be Allocated in 2024-2025





Thesis Point 2 – Increasing demand for energy, including for AI applications, will increase demand for nuclear energy sources

Second Thesis Point

Increasing demand for energy, including for AI applications, will increase demand for nuclear energy sources

Increasing Demand for Energy

■ Data Growth and Investment Trends

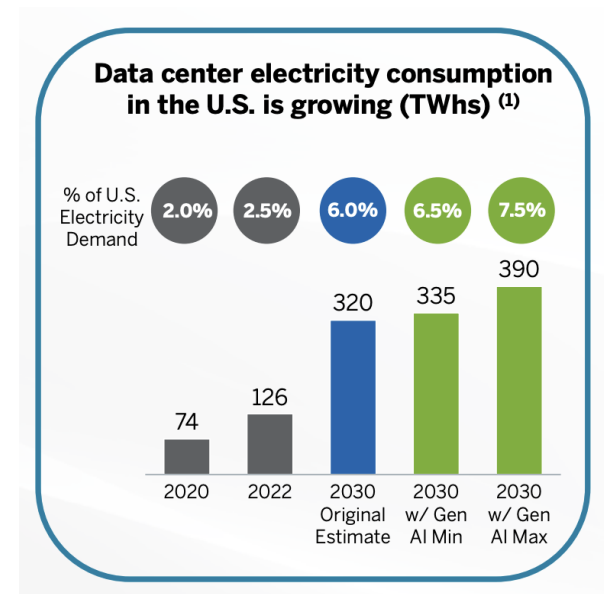
- *Data Explosion:* Data output will double in the next five years (CEG Investor Slides 9/24)
- *Investment Surge:* \$1 trillion in data center investments is projected over the same period

■ Power Demands of AI

- *AI Racks' Power Needs:* AI racks use up to 7x more power than traditional ones.
- *Investment Continues:* Same \$1 trillion investment emphasizes AI infrastructure growth

■ Projected Energy Impact:

- *IEA Forecast:* Data centers could drive one-third of U.S. electricity demand growth from 2024-2026



Second Thesis Point Continued

Increasing demand for energy, including for AI applications, will increase demand for nuclear energy sources

Strategic Nuclear Partnerships

- **Nvidia Supports Nuclear Power:**

- *Nvidia's Stance:* CEO backs nuclear as essential for AI data centers

- **Microsoft's Nuclear Deal:**

- *20-Year Contract:* Microsoft partners with Constellation Energy for nuclear power

- **Why Nuclear Energy?:**

- *Reliable & Low-Carbon:* Consistent energy for AI, with a low carbon footprint





Thesis Point 3 – Constellation is Ideally Positioned to Capture Growing Demand in the Clean Energy Market

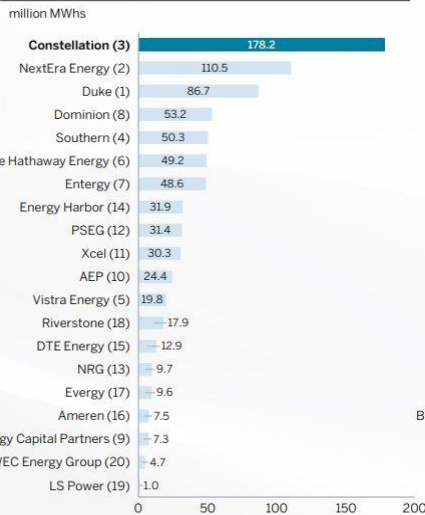
Third Thesis Point

Ideally Positioned to Capture Emerging Market Demand

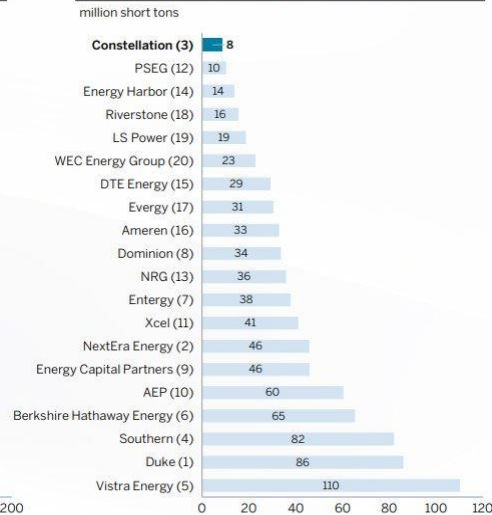
Industry Influence

- #1 producer of carbon-free energy in the United States
 - At the forefront of a growing industry
- Manages \$50.76 Billion in Assets (2023)
 - Capital intensive nature of industry deters competition
 - Continued infrastructure investment demonstrates the ability to service emerging clean energy demand

Largest Producers of Carbon-Free Generation (1,2)



Lowest CO₂ Emissions Among Major Investor-Owned Generators (2)



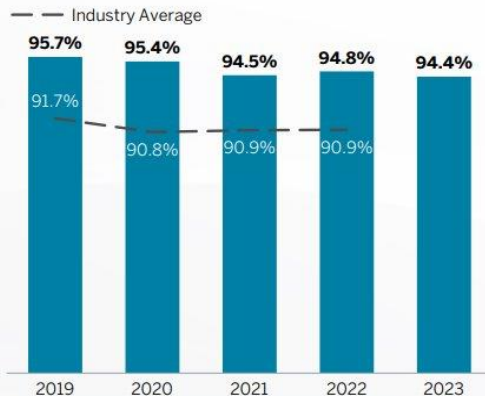
Third Thesis Point Continued

Ideally Positioned to Capture Emerging Market Demand

Unmatched Production Efficiency

- #1 among competitors in nuclear capacity factor (actual output/maximum potential output)
 - Lowest industry average maintenance outage days for core refueling
- Federal Production Tax Credit Through Year-End 2032
 - Subsidized industry shows government support, encourages growth and provides baseline earnings stability

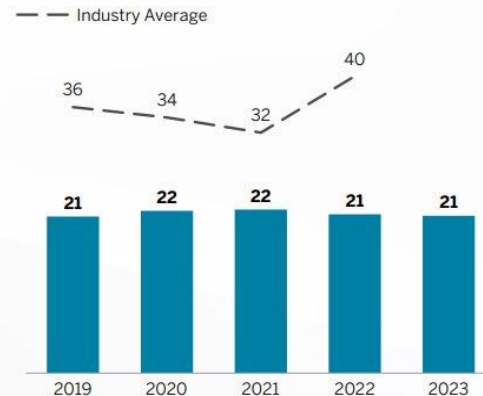
1-Yr Nuclear Capacity Factor (%) ^(1,2)



Ranking Among Major Operators (2-Yr) ⁽³⁾

2019	2020	2021	2022
1	1	1	1

Average Nuclear Refueling Outage Days ^(2,4)



Ranking Among Major Operators ⁽³⁾

2019	2020	2021	2022
1	1	1	1

Valuation:

Company	Ticker	Market Cap	Enterprise Value	Revenue		EBITDA		NTM Multiples	
				LTM	NTM	LTM	NTM	EV/ Revenue	EV/ EBITDA
The Southern Company	SO	95,784.7	163,455	26,428	27,728	13,061	11,588	5.9x	14.1x
PG&E Corporation	PCG	43,283.6	102,831	24,829	25,455	9,318	4,081	4.0x	25.2x
Dominion Energy	D	47,939.3	90,556	14,593	15,906	7,032	5,298	5.7x	17.1x
Duke Energy Corporation	DUK	87,745.3	173,790	29,751	30,913	14,106	4,262	5.6x	40.8x
Excelon Corporation	EXC	38,505.2	84,004	22,924	23,850	7,148	8,718	3.5x	9.6x
Constellation Energy	CEG	73,497	80,487	25,369.5	20,464	23,982	4,874	3.9x	16.5x
Low								3.5x	9.6x
Mean								5.0x	21.4x
Median								5.6x	17.1x
High								5.9x	40.8x

Note: Values are in millions(mm)

NTM values calculated based on quarter projected earnings

Valuation continued

Bear Case:

Value of CEG stock is contingent on increases in renewable energy deals. If negotiations on these deals are halted, or even slowed, CEG's price will take a huge hit. That is, Implied ebitda is miscalculated and the stock is currently mispriced.

Base Case:

Being a utilities company, CEG will be able to weather recession conditions and still be a profitable company. Even if deal growth were to slow, CEG has a solid portion of market share and reliable revenues. Implied EBITDA growth is higher than average, but the stock will still be priced efficiently.

Bull Case:

With high correlations to AI growth and reliance on Nuclear energy to power that AI, the EBITDA implied by CEG's current price is too low. This mispricing allows for serious upside potential.

Note: Values calculated on 3-year growth estimates

EV/EBITDA	Implied Share Price	Implied return
1.8x	365.09	53%
1.6x	265.99	11%
1.3x	154.00	-36%

Constellation Energy Group

- Supplier Bargaining Power: **Low to Moderate**
 - CEG has multiple suppliers for its fuel sources giving any individual supplier low power
- Buyer Bargaining Power: **Moderate**
 - Many consumers don't have much choice in who supplies their energy
- Threat of New Entrants: **Low**
 - High initial start-up costs and strong regulations limit entry
- Threat of Substitution: **High**
 - Many alternative clean energy sources such as solar and wind take market share
- Competitive Rivalry: **Moderate**
 - Few firms but strong competition because there is no product differentiation

Constellation Energy Group

- Total Score - 28.5 - Medium
- Environmental
 - Powers 16 Million Homes Carbon Free
 - Avoids 125 Million Metric Tons of CO2
 - Ranked the 9th Most Sustainable Company in the US
- Social
 - Diversity Equity and Inclusion Top Scorer
 - 100,000 Hours Volunteered by Employees
- Governance
 - Regulatory Risk
 - Spent Nuclear Fuel Storage

QUESTIONS?